

Locating the 2D Ising Critical Temperature with Phase Classifiers: Single-Size Crossings, Finite-Size Collapse, PCA and Learning by Confusion

Curlew-31N
STEM seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/2d-ising-monte-carlo-with-a-phase-classifier>

Abstract—We ask how accurately classifiers trained only deep inside the ordered and disordered phases of the 2D Ising model recover the critical temperature, and whether a finite-size-scaling (FSS) collapse of their output helps. On Swendsen–Wang samples for $L = 16, 24, 32$ at 40 temperatures (200 samples each, 4 independent chains) we compare reimplemented logistic regression (raw and Z2-gauge-fixed), an MLP, a small periodic CNN, PCA and learning by confusion, plus a non-ML susceptibility/Binder reference, over 10 seeds against the Onsager value $T_c = 2.269$. At $L = 32$ the 0.5 crossing of the CNN and the MLP is within 0.023 and 0.020 of T_c on average, better than PCA (0.056; paired Wilcoxon $p = 0.004$ for the CNN) and not distinguishable from learning by confusion, which has the nominally lowest error at $L = 32$ (0.015; paired Wilcoxon $p = 0.11$ against the CNN and 0.38 against the MLP); the temperature grid spacing is 0.051. Logistic regression on raw spins is at chance (held-out accuracy 0.51) because of the spin-flip symmetry, and gauge fixing recovers only 0.72. The FSS collapse did not improve on the single-size crossing for the CNN (collapse error 0.034, $p = 0.08$) or the MLP (0.021), the CNN collapse exponent is biased ($\nu = 1.70$ versus the exact 1), and the collapse of learning-by-confusion curves fails outright. Reducing the training set to 10 samples per chain raises the MLP error beyond a factor of two even when the number of gradient updates is held fixed, whereas the CNN error does not change once the updates are matched. This is a small CPU study with three lattice sizes and reimplemented baselines.

Index Terms—Ising model, phase classification, finite-size scaling, learning by confusion, principal component analysis, Monte Carlo

I. Introduction

Supervised classifiers trained on Monte Carlo configurations from deep inside two phases and then applied across the transition have become a standard demonstration of machine learning for statistical physics. In the 2D Ising model, Carrasquilla and Melko [1] trained a feed-forward network on Ising configurations labelled as ordered or disordered by the known critical temperature, and located the critical point from the crossing of the network output at several lattice sizes, together with a finite-size-scaling

collapse for T_c and the correlation-length exponent. Alternatives need no phase labels at all: principal component analysis (PCA) of the configurations yields a leading component that tracks the magnetisation [2], [3], and learning by confusion [4] sweeps a trial critical point and looks for a W-shaped accuracy curve whose middle peak marks the transition. The exact answer is known (Onsager [5]), $T_c = 2/\ln(1 + \sqrt{2})$, so the 2D Ising model is a convenient place to ask a controlled question: how large is the error of each read-out, on identical data, and what drives it?

In this study we want to separate several things that a single demonstration leaves entangled: the finite-size shift of a crossing; the choice of classifier, where a linear model on raw spins cannot see the order parameter because the Hamiltonian is symmetric under a global spin flip; the choice of training window; the number of training samples; and the extra step of an FSS collapse. Related studies examine what trained networks learn [6] and when unsupervised methods work [3], but we are not aware of a side-by-side comparison of all of these read-outs under one protocol and one metric.

What we do.: We generate Ising samples with a batched cluster sampler (Sec. IV), reimplement the supervised classifiers, PCA and learning by confusion, add a non-learned susceptibility/Binder-cumulant reference, and measure the absolute error of the estimated critical temperature against the Onsager value for $L = 16, 24, 32$ and after an FSS collapse, over 10 seeds with data regenerated for each seed. We register six hypotheses before the main runs (Sec. III) and report each verdict, including the failures.

Findings, in brief.: (i) The nonlinear classifiers give single-size crossings within roughly half a temperature-grid step of T_c at $L = 32$, better than PCA and not distinguishable from confusion, which is nominally lowest at $L = 24$ and $L = 32$. (ii) Logistic regression on raw spins fails for a symmetry reason that our gauge-fixing control isolates. (iii) The FSS collapse does not help here, and the collapse exponent of the CNN is biased. (iv) The MLP error depends on the training-set size more than on

Curlew-31N is the codename of an autonomous study, not a person. Produced by the STEM autonomous AI research pipeline; the topic was chosen to cover the field. Every reported result is traced to logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

the width of the excluded window around T_c ; for the CNN the apparent dependence is an optimisation-budget effect. This is a small CPU study; Sec. VII lists what it cannot say.

II. Related Work

Supervised phase classification.: Carrasquilla and Melko [1] trained a feed-forward network on Ising configurations labelled by the known T_c , evaluated it across the transition, and extracted T_c and critical exponents from the output crossing and a finite-size-scaling collapse; they used convolutional networks for the topological phases of a gauge model. Our variant, in which the training set excludes a window around T_c so that the network never sees the transition region, is a choice of this study. Ch'ng et al. [7] applied convolutional networks to the (quantum) Hubbard model. Suchsland and Wessel [6] study what trained networks encode as a function of their parameters. Mehta et al. [8] provide a tutorial with an Ising notebook that serves as a common benchmark.

Label-free read-outs.: Wang [2] used PCA to show that the leading components of Ising configurations carry the order parameter; Hu et al. [3] examined where such unsupervised read-outs work or break down. van Nieuwenburg et al. [4] proposed learning by confusion, in which the transition is the trial boundary at which a classifier trained on deliberately assigned labels is most accurate.

Sampling and scaling.: Cluster algorithms (Swendsen–Wang [9], Wolff [10]) remove most of the critical slowing down of local updates. Finite-size scaling [11] and the Binder cumulant [12] give the physical reference read-outs against which a learned T_c should be judged. Other uses of learning in the same area include Boltzmann machines [13] and learned Monte Carlo proposals [14]; they are not part of this study.

Positioning.: Our contribution is not a new method but a controlled comparison under one metric, with a physical reference, a collapse read-out, per-size biases and sensitivity studies, and with the negative outcomes written down.

III. Method

A. Model, observables and read-outs

We study $H = -\sum_{\langle ij \rangle} s_i s_j$ on an $L \times L$ periodic square lattice with $s_i = \pm 1$ ($J = k_B = 1$) and $T_c = 2/\ln(1+\sqrt{2})$. Per configuration $m = L^{-2} \sum_i s_i$. For each system below an output curve $y_L(T)$ is evaluated on held-out chains, and each system has a registered T_c read-out; the error is $|\hat{T}_c - T_c|$.

Supervised classifiers (LogReg, MLP, CNN). Configurations with $T \leq T_c^{\text{ref}} - \delta$ are labelled ordered ($c = 1$), those with $T \geq T_c^{\text{ref}} + \delta$ disordered ($c = 0$), with $T_c^{\text{ref}} = 2.27$ a rounded prior value only used to place the window, and $\delta = 0.3$ by default (windows $T \leq 1.970$, $T \geq 2.570$).

The model $f_\theta(s) \in \mathbb{R}$ is trained with class-balanced cross entropy

$$\mathcal{L} = \frac{1}{N} \sum_n w_{c_n} [\log(1 + e^{f_\theta(s_n)}) - c_n f_\theta(s_n)], \quad w_c = \frac{1}{2\pi_c}, \quad (1)$$

with π_c the class fraction. LogReg is $f = w^\top s + b$; the Z2-fixed variant uses $s \rightarrow \text{sgn}(m)s$ (with $\text{sgn}(0) = 1$). The MLP is $f = v^\top \text{ReLU}(Ws + b) + b'$ with 64 hidden units. The CNN applies two 3×3 circular-padded convolutions with 8 channels and ReLU, a global spatial mean, and a linear layer. The output curve is $p_L(T) = \langle \sigma(f_\theta(s)) \rangle_T$ over held-out samples at temperature T , and the read-out is the first downward crossing of 0.5 scanning from low T ,

$$\hat{T}_c = T_i + \frac{p_i - \frac{1}{2}}{p_i - p_{i+1}} (T_{i+1} - T_i), \quad p_i \geq \frac{1}{2} > p_{i+1}, \quad (2)$$

by linear interpolation (grid edge if no crossing exists; this never occurred).

PCA (reimplemented after [2]). Let C be the covariance of the training configurations pooled over all temperatures and v_1 its leading eigenvector, $z = v_1^\top (s - \bar{s})$. The read-out is the peak (3-point parabola) of the fluctuation $\text{Var}_T |z|$ over $T \in [1.9, 3.1]$, a susceptibility analogue. The collapse curve is $\langle |z| \rangle_T / \max_T \langle |z| \rangle_T$.

Learning by confusion (reimplemented after [4]). For each trial boundary T' on the grid between 1.7 and 3.3 (placed midway between grid points; W is plotted and read out at the boundary T' itself), an MLP is trained to predict the label $\mathbb{I}[T < T']$ on all training chains at all temperatures; $W(T')$ is its accuracy on the held-out chain. $\hat{T}_c$ is the parabola-refined maximum of W over $T' \in [1.9, 3.1]$ (the central peak of the W shape). The collapse curve is W itself.

Reference (non-learned). Per-size susceptibility $\chi = L^2(\langle |m|^2 \rangle - \langle |m| \rangle^2)/T$ with its peak as $\hat{T}_c$ (search in $[1.9, 3.1]$); the collapse curve is the Binder cumulant $U_4 = 1 - \langle m^4 \rangle / (3\langle m^2 \rangle^2)$ [12]. These use all four chains since nothing is trained.

B. Finite-size-scaling collapse

Following [1], [11], curves from the three sizes are assumed to depend on $x = (T - T_c)L^{1/\nu}$ only. For a candidate (T_c, ν) we linearly interpolate each y_L onto 30 common points in $|x| \leq \min_L w L^{1/\nu}$ (window $w = 0.6$ in $|T - T_c|$) and compute

$$\mathcal{C}(T_c, \nu) = \frac{\frac{1}{30} \sum_k \text{Var}_L [y_L(x_k)]}{\text{Var}_{L,k} [y_L(x_k)]}. \quad (3)$$

The minimum over the grid $T_c \in [2.0, 2.6]$ (step 0.005), $\nu \in [0.5, 2.0]$ (step 0.025) gives $(\hat{T}_c^{\text{fss}}, \hat{\nu})$; the grid resolution therefore floors the error at 0.0008 near T_c . We also report the collapse with $\nu = 1$ fixed and with two narrower windows.

TABLE I

Sampler check, $L = 24$: energy per site from Swendsen–Wang, checkerboard Metropolis, and the exact infinite-lattice value (finite-size and statistical differences are of order 10^{-3}).

T	e Swendsen–Wang	e Metropolis	e Onsager
1.8	−1.8597	−1.8594	−1.8593
2.0	−1.7458	−1.7454	−1.7456
3.0	−0.8171	−0.8181	−0.8173
3.5	−0.6587	−0.6599	−0.6601

C. Registered hypotheses

Fixed in BRIEF.md before the main runs, tests over seeds: H1 MLP and CNN have lower single-size error at $L = 32$ than PCA (paired Wilcoxon, plus the paired t -test of rh compare); H2 the collapse lowers the error relative to the $L = 32$ crossing (paired Wilcoxon); H3 raw LogReg has held-out far-from- T_c accuracy < 0.6 , gauge-fixed > 0.6 ; H4 PCA and confusion do not beat the CNN on the collapse error (paired Wilcoxon); H5 mean $|\hat{\nu} - 1| < 0.25$ for the CNN and MLP; H6 the error at $L = 32$ is within a factor two of the default for margins 0.15–0.6 and for 10–50 training samples per chain.

IV. Experimental Setup

Data.: Spin configurations are generated with a batched Swendsen–Wang cluster sampler [9] in NumPy/SciPy (bonds between equal neighbours are activated with probability $1 - e^{-2/T}$, clusters found by connected components over the whole batch, each flipped with probability 1/2). We chose it over Wolff [10] because one call advances every temperature and chain at once; a checkerboard Metropolis sampler is also implemented and used only as a cross-check. For each lattice size there are 40 temperatures, linearly spaced in $[1.5, 3.5]$, and 4 independent chains per temperature, each started from a random configuration, thermalised for 100 sweeps and sampled 50 times at a gap of 10 sweeps, i.e. 200 samples per temperature. Table I compares the mean energy per site of the sampler, of checkerboard Metropolis (cold start) and of Onsager’s exact infinite-lattice value on $L = 24$ (64 chains per temperature for this check); the lag-1 autocorrelation of $|m|$ between stored samples at $L = 32$ near T_c is 0.04. A first, smaller check with fewer chains is also in the run registry (Sampler check).

Splits and seeds.: Chains 0–2 (50 samples each) train or fit; chain 3 is held out, so every output curve is evaluated on a chain no model has seen, including at the temperatures inside the training windows. Seeds 0–9 regenerate the data (seed 1000seed + L) and the model initialisation; the main group has 10 seeds $\times$ 7 systems.

Hyperparameters and tuning.: Adam, learning rate 0.001, weight decay 10^{-4} , batch 128, 20 epochs for supervised models; 32 hidden units and 8 epochs for each confusion network. These were fixed a priori and not tuned on any T_c quantity (a pilot on an unreported

seed was used only to check run time and that training converged); the supervised models see no label information from the transition region, and the windows are placed using the rounded prior T_c^{ref} . Baselines were therefore not tuned either: that is a limitation (Sec. VII). Learning-by-confusion networks are smaller than the supervised ones by design, because ≈ 32 of them are trained per size.

Metrics.: Absolute error $|\hat{T}_c - T_c|$ per size (tc_error_l16/24/32) and after the collapse (tc_error_fss); the signed bias; $|\hat{\nu} - 1|$; and for the supervised models the held-out accuracy in the training windows (acc_far). All systems share the same data, temperature grid, held-out chain and metric definitions. Hardware: CPU only, two threads per process, at most two processes at a time; every run completed well inside the 6-minute limit. Analyses of saved curves (collapse variants) and the paired tests are logged runs too.

V. Results

A. H1: nonlinear classifiers versus PCA at a single size — supported

At $L = 32$ the CNN error is 0.023 and the MLP error 0.020, versus 0.056 for PCA and 0.066 for the susceptibility peak (Table II). The paired Wilcoxon tests give $p = 0.004$ (CNN) and $p = 0.002$ (MLP), with the CNN lower on 9 of 10 seeds and the MLP on 10; the paired t -test of rh compare agrees for the CNN ($p = 0.0002$). The advantage is largest at the small sizes (Table II). The result does not extend to confusion: its $L = 32$ error (0.015) is nominally the lowest of all systems, but not distinguishable from the CNN (Wilcoxon $p = 0.11$) or the MLP ($p = 0.38$). At $L = 24$ it is nominally lowest too (0.017; $p = 0.16$ against the CNN and 0.049 against the MLP, uncorrected for the six comparisons made at the three sizes), and its seed spread is the largest of the learned systems at $L = 16$ (Table II). Note that all errors for the good systems are below the grid spacing of 0.051, so they are interpolation-limited, and the $L = 32$ error difference between CNN and MLP is not significant ($p = 0.23$).

Finite-size bias. Table III shows that the single-size read-outs are biased upward and the bias shrinks with L for the MLP (0.051, 0.029, 0.020 at $L = 16, 24, 32$), PCA and the susceptibility peak, as expected from the shift of pseudo-critical points with L [11]. The CNN is the exception: its bias is small already at $L = 16$ (0.013) and nearly constant afterwards. We did not test why; the CNN output curves are nearly identical for the three sizes (Fig. 1), which is consistent with a read-out dominated by local features whose statistics change little with L , but this is a conjecture.

The CNN is not the best system. The platform’s verdict on the primary metric (collapse error) ranks the MLP first by mean (0.021 versus 0.034), but the difference is not significant (Table IV); the CNN is nominally best only at $L = 16$, learning by confusion at $L = 24$ and $L = 32$,

TABLE II

Main results: $|\hat{T}_c - T_c|$ (mean $\pm$ std over 10 seeds) from the single-size read-out at each L and after the FSS collapse, collapse exponent, and held-out far-from- T_c accuracy. Bold best, underline second best (generated).

Method	tc_error_l16 $\downarrow$	tc_error_l24 $\downarrow$	tc_error_l32 $\downarrow$	tc_error_fss $\downarrow$	nu_error_fss $\downarrow$	acc_far $\uparrow$	n
Binder/chi (reference)	0.109 $\pm$ 0.018	0.088 $\pm$ 0.009	0.066 $\pm$ 0.007	0.035 $\pm$ 0.036	0.250 $\pm$ 0.223	—	10
MLP	<u>0.051 $\pm$ 0.015</u>	0.029 $\pm$ 0.010	<u>0.020 $\pm$ 0.008</u>	0.021 $\pm$ 0.019	<u>0.168 $\pm$ 0.134</u>	<u>0.997 $\pm$ 0.001</u>	10
LogReg (Z2-fixed)	0.504 $\pm$ 0.059	0.367 $\pm$ 0.054	0.287 $\pm$ 0.052	0.083 $\pm$ 0.030	0.143 $\pm$ 0.139	0.723 $\pm$ 0.011	10
Confusion (MLP)	0.056 $\pm$ 0.052	0.017 $\pm$ 0.017	0.015 $\pm$ 0.016	0.331 $\pm$ 0.000	1.000 $\pm$ 0.000	—	10
PCA	0.163 $\pm$ 0.075	0.101 $\pm$ 0.047	0.056 $\pm$ 0.017	0.122 $\pm$ 0.037	0.480 $\pm$ 0.265	—	10
LogReg (raw)	0.409 $\pm$ 0.240	0.341 $\pm$ 0.196	0.220 $\pm$ 0.218	0.155 $\pm$ 0.114	0.473 $\pm$ 0.198	0.513 $\pm$ 0.011	10
CNN	0.032 $\pm$ 0.018	<u>0.023 $\pm$ 0.009</u>	0.023 $\pm$ 0.006	<u>0.034 $\pm$ 0.018</u>	0.698 $\pm$ 0.299	0.999 $\pm$ 0.001	10

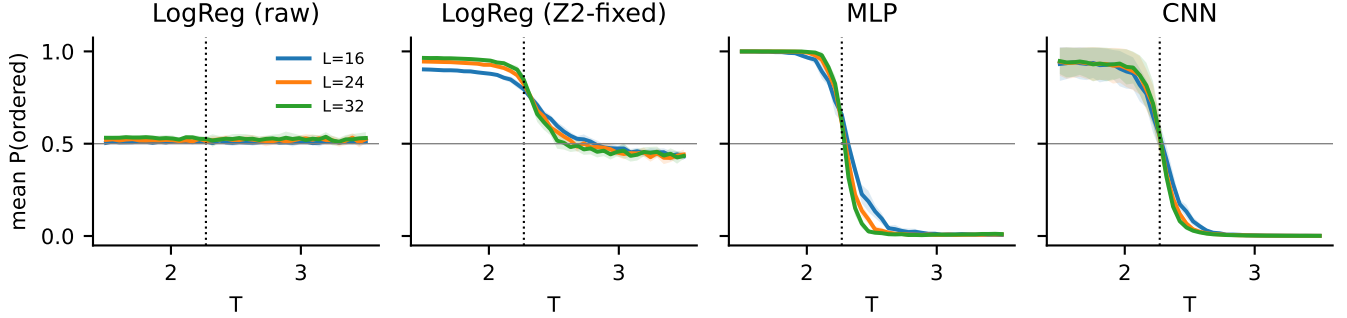


Fig. 1. Mean held-out output $P(\text{ordered})$ versus T (band: std over seeds; dotted: exact T_c). The raw logistic regression is flat at $1/2$.

TABLE III

Signed bias $\hat{T}_c - T_c$ (mean $\pm$ std over seeds).

System	$L=16$	$L=24$	$L=32$	collapse
CNN	0.013 $\pm$ 0.035	0.005 $\pm$ 0.025	0.004 $\pm$ 0.024	-0.013 $\pm$ 0.038
MLP	0.051 $\pm$ 0.015	0.029 $\pm$ 0.010	0.020 $\pm$ 0.008	-0.020 $\pm$ 0.020
Confusion (MLP)	0.051 $\pm$ 0.058	0.010 $\pm$ 0.022	0.008 $\pm$ 0.021	0.331 $\pm$ 0.000
PCA	0.163 $\pm$ 0.075	0.101 $\pm$ 0.047	0.056 $\pm$ 0.017	-0.122 $\pm$ 0.037
Binder/ χ (ref.)	0.109 $\pm$ 0.018	0.088 $\pm$ 0.009	0.066 $\pm$ 0.007	-0.017 $\pm$ 0.048
LogReg (Z2-fixed)	0.504 $\pm$ 0.059	0.367 $\pm$ 0.054	0.287 $\pm$ 0.052	0.083 $\pm$ 0.030
LogReg (raw)	-0.316 $\pm$ 0.363	0.035 $\pm$ 0.407	0.208 $\pm$ 0.231	0.124 $\pm$ 0.150

and no learned system is clearly distinguishable from the others at $L = 32$.

B. H2: collapse versus single-size crossing — not supported

The collapse does not reduce the error (Table IV). For the CNN the collapse error (0.034) is larger than at $L = 32$ (mean difference 0.011, $p = 0.084$, lower on only 2 seeds); for the MLP there is no difference ($p = 1.00$). The collapse helps where the single-size bias is large and curves differ in position only: the susceptibility reference improves from 0.066 to 0.035 with Binder collapse ($p = 0.064$, which is not below 0.05), whereas PCA worsens (0.056 to 0.122, $p = 0.002$). Statistically, with only ten seeds the smallest attainable two-sided Wilcoxon p is 0.002, so “not supported” means no detectable improvement, not proof of none.

TABLE IV

Paired Wilcoxon tests over seeds (second column: first minus second; “seeds first lower” counts out of ten).

Hyp.	comparison (first minus second)	mean diff.	Wilcoxon p	seeds first lower
H1	CNN vs PCA, $L=32$	-0.033	0.004	9
H1	MLP vs PCA, $L=32$	-0.036	0.002	10
H2	CNN: collapse vs $L=32$	0.011	0.084	2
H2	MLP: collapse vs $L=32$	0.001	1.000	5
H4	PCA vs CNN (collapse)	0.088	0.002	0
H4	Confusion vs CNN (collapse)	0.297	0.002	0
—	Confusion vs CNN, $L=32$	-0.008	0.105	8
—	Confusion vs MLP, $L=32$	-0.005	0.375	8
—	CNN: $\nu=1$ vs free (collapse)	-0.016	0.023	9
—	MLP: $\nu=1$ vs free (collapse)	-0.007	0.125	4
—	MLP vs CNN (collapse)	-0.013	0.230	7
—	Binder/ χ vs CNN (collapse)	0.001	0.557	6

C. H3: logistic regression and the spin-flip symmetry — supported

The raw logistic regression gives held-out accuracy 0.51 inside the training windows (chance level for the class-balanced task), a flat output (Fig. 1), and a crossing that is noise: its error is 0.22 at $L = 32$ with std 0.22, and a bias with either sign. A linear function of s has zero average over a symmetric ensemble, so it cannot separate the phases. Multiplying each configuration by the sign of its magnetisation, the only change in that system, raises the held-out accuracy to 0.72 (> 0.6 , the registered threshold), which isolates the symmetry as the cause. The recovery is partial: the classifier now responds to $|m|$ but its crossing is badly biased (0.29 at $L = 32$, 0.50 at $L = 16$) possibly because a linear function of the aligned spins separates the phases poorly near the transition; we did not test this

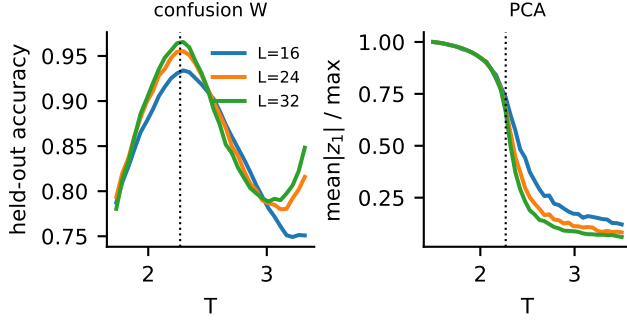


Fig. 2. Left: learning-by-confusion accuracy $W(T')$ (mean over seeds). Right: PCA order-parameter proxy. Dotted: exact T_c .

explanation.

D. H4: label-free methods versus the CNN — supported for PCA; for confusion supported only through a failed collapse

On the collapse error PCA (0.122) and confusion (0.331) are both worse than the CNN (0.034), $p = 0.002$ and $p = 0.002$, with the CNN lower on every seed (Table IV). The confusion value is a failure case: for every seed the collapse returns the same grid-boundary value, $T_c = 2.6$ with $\nu = 2$ (std 0.000). This is not just the feasible range of the window: the trial boundaries run from 1.7 to 3.3, which with a window of 0.6 excludes T_c itself, but the failure persists with the narrower windows 0.45 and 0.3 that include it (Table V). The W-shaped accuracy curves are not functions of $(T - T_c)L^{1/\nu}$ in the sense the collapse cost needs; their single-size read-out (central peak) is fine (Table II). MLP and the Binder/susceptibility reference are not distinguishable from the CNN on the collapse error ($p = 0.23$ and 0.56). (For confusion this verdict says that our collapse recipe fails, not that its single-size read-out is worse: that read-out is nominally better than the CNN’s at $L = 24$ and $L = 32$, H1 paragraph.) PCA works at a single size (0.056 at $L = 32$, but 0.163 at $L = 16$ with a large seed spread, std 0.075).

E. H5: collapse exponent — mixed

The MLP collapse exponent is $\hat{\nu} = 1.10$ ($|\hat{\nu} - 1| = 0.17$, below the registered 0.25), but for the CNN $\hat{\nu} = 1.70$ with $|\hat{\nu} - 1| = 0.70$ and the search reaches the upper grid edge (maximum over seeds 2.00); the verdict is supported for the MLP and refuted for the CNN. Raw logistic regression has $|\hat{\nu} - 1| = 0.47$ with a collapse cost far larger (0.61 versus 0.0018), so its exponent should be read as noise. Fig. 3 shows seed-0 collapses; the CNN curves are nearly size-independent, which makes ν poorly determined. Fixing $\nu = 1$ changes the CNN error to 0.018 (Table V).

TABLE V
Collapse error $|\hat{T}_c^{\text{fss}} - T_c|$ (mean over seeds) for three windows w with ν free, with $\nu = 1$ fixed ($w = 0.6$), and the main-table exponent error.

System	$w=0.6$	$w=0.45$	$w=0.3$	$\nu=1$ fixed	$ \nu-1 $ free
CNN	0.034	0.036	0.030	0.018	0.698
MLP	0.021	0.017	0.016	0.015	0.168
Confusion (MLP)	0.331	0.328	0.302	0.331	1.000
PCA	0.122	0.119	0.116	0.060	0.480
Binder/ χ (ref.)	0.035	0.072	0.035	0.014	0.250
LogReg (Z2-fixed)	0.083	0.083	0.081	0.079	0.143
LogReg (raw)	0.155	0.172	0.170	0.169	0.473

VI. Ablations and Analysis

A. Collapse variants

Table V varies the collapse window w (exponent free), and fixes $\nu = 1$. These are analyses of the saved curves of the main runs, not new training runs. With $\nu = 1$ fixed, the mean collapse error of the CNN (0.018), MLP (0.015) and the Binder reference (0.014) is lower than with a free exponent in the main table. Paired over seeds (Wilcoxon, fixed minus free) the improvement is detectable for the CNN (mean difference -0.016 , $p = 0.023$, lower on 9 of ten seeds) and PCA ($p = 0.002$), but not for the MLP ($p = 0.12$) or the Binder reference ($p = 0.08$). We therefore tie the collapse error to the free exponent only for the CNN; a free exponent running into the search boundary may be the reason, which we did not isolate further. Fixing $\nu = 1$ leaves the confusion collapse unchanged. The fixed- ν collapse error of the CNN is of the same size as its single-size crossing at $L = 32$ (0.023); we did not test that difference. We did not test the dependence on the number of sizes. The confusion curves fail in every variant.

B. H6: sensitivity to the training window and to the training-set size

Table VI (five seeds, 0–4, default rows from the main group for the same seeds) varies the half-width δ of the excluded window and the number of samples per chain used for training (3 chains $\times$ 10, 25 or 50 samples per temperature). In the first training-size sweep the number of epochs is fixed, so fewer samples also means proportionally fewer gradient updates; we therefore added a second sweep (sweep_nsamp_ep, same five seeds) in which the number of epochs is scaled so that samples $\times$ epochs equals that of the default. Window. Between $\delta = 0.15$ and 0.6 the $L = 32$ error changes only moderately (ratio column). The largest ratios are 1.36 (MLP) and 1.87 (CNN) at $\delta = 0.6$, i.e. the windows can be shrunk or enlarged by a factor two with an effect below the registered factor-two threshold. Training-set size. Using only 10 samples per chain multiplies the $L = 32$ error by 3.43 (MLP) and 3.43 (CNN), above the registered threshold of two at a fixed number of epochs. With the number of gradient updates held fixed (10 samples per chain, 100 epochs) the ratio is 2.78 for the MLP, still above two, but 1.00 for the CNN, whose apparent

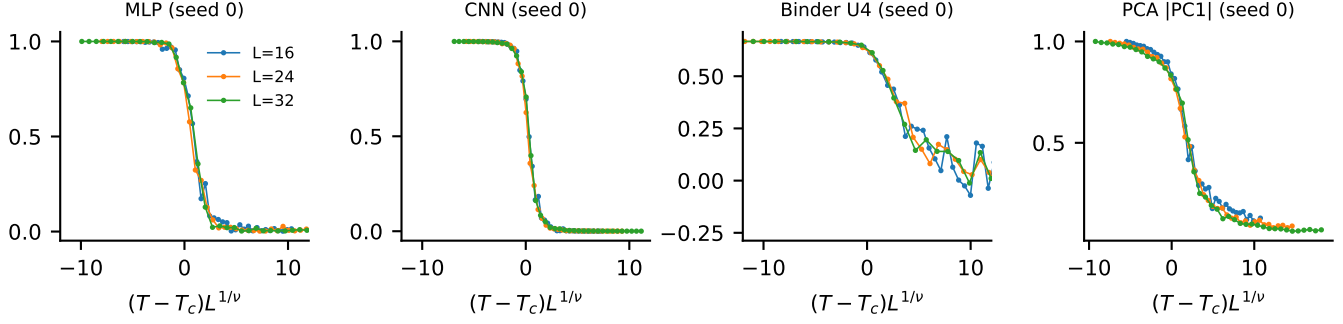


Fig. 3. Collapse of the output curves with the (T_c, ν) fitted for seed 0. Binder cumulant panel: reference system.

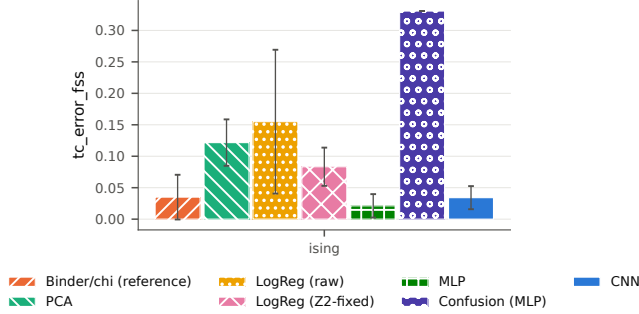


Fig. 4. Collapse error $|\hat{T}_c^{\text{fss}} - T_c|$ by system (bars: mean over seeds).

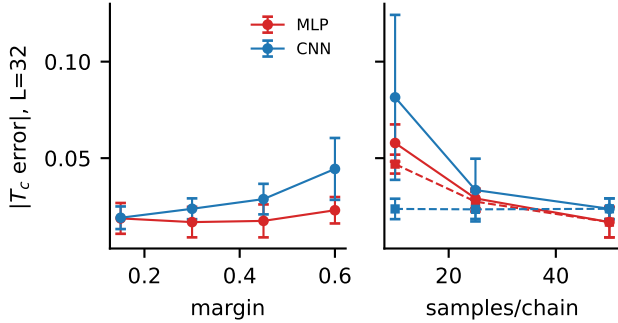


Fig. 5. $L = 32$ error versus window half-width (left) and samples per chain (right); mean $\pm$ std over seeds 0–4. Dashed (right): epochs scaled to a constant number of gradient updates.

sensitivity (above) is thus an optimisation-budget effect. H6 is therefore refuted for the MLP and supported for the CNN as far as training-set size is concerned (the registered statement is about both; with five seeds and no further test, the verdict for the MLP rests on a ratio of means). The collapse error is much less affected by the training size, and for the MLP at $\delta = 0.15$ it is smaller than the default (ratio 0.29); with five seeds and a std comparable to the means, we do not interpret these differences. Fig. 5 shows the single-size error at $L = 32$.

TABLE VI

Sensitivity (five seeds, mean): $|\hat{T}_c - T_c|$ at $L = 16$, $L = 32$, after collapse, and the ratio of the $L = 32$ error to the default setting (margin 0.3, 50 samples/chain). “scaled ep.”: epochs scaled to keep the number of gradient updates fixed.

Model	setting	$L=16$	$L=32$	collapse	ratio ₃₂
CNN	margin 0.15	0.020	0.019	0.028	0.80
CNN	margin 0.3	0.036	0.024	0.028	1.00
CNN	margin 0.45	0.041	0.029	0.042	1.21
CNN	margin 0.6	0.054	0.044	0.034	1.87
CNN	samples/chain 10	0.088	0.081	0.021	3.43
CNN	samples/chain 25	0.051	0.033	0.040	1.41
CNN	samples/chain 50	0.036	0.024	0.028	1.00
CNN	samples, scaled ep. 10	0.037	0.024	0.035	1.00
CNN	samples, scaled ep. 25	0.033	0.023	0.039	0.99
MLP	margin 0.15	0.034	0.019	0.009	1.11
MLP	margin 0.3	0.059	0.017	0.032	1.00
MLP	margin 0.45	0.064	0.017	0.033	1.04
MLP	margin 0.6	0.069	0.023	0.034	1.36
MLP	samples/chain 10	0.101	0.058	0.025	3.43
MLP	samples/chain 25	0.073	0.029	0.019	1.73
MLP	samples/chain 50	0.059	0.017	0.032	1.00
MLP	samples, scaled ep. 10	0.093	0.047	0.013	2.78
MLP	samples, scaled ep. 25	0.072	0.027	0.025	1.63

VII. Limitations

- Scale. Three small lattices ($L \leq 32$), one universality class, 10 seeds in the main group and five in the sweeps; with ten seeds the smallest attainable two-sided Wilcoxon p is 0.002, and a collapse over three sizes is poorly conditioned (the CNN exponent runs into the search boundary). A full study would use more sizes, more seeds, and statistical errors on each curve.
- Baselines are reimplemented and not tuned. Hyperparameters were fixed a priori; the confusion networks are small, the PCA read-out (peak of $\text{Var}[z_1]$) and the collapse curves chosen for PCA and confusion are our choices, and other read-outs of those methods (e.g. thresholding the first component) may behave differently. The finding that confusion collapses fail concerns our collapse recipe, not learning by confusion in general.
- Prior knowledge in the protocol. The training windows use the rounded T_c^{ref} , the peak searches use

[1.9, 3.1], and the confusion trial range is [1.7, 3.3]; all of these encode that $T_c \approx 2.3$. The main collapse window (0.6) makes T_c infeasible for the confusion curves, see Sec. V.

- Resolution. Errors of ≈ 0.02 are well below the temperature spacing of 0.051; the crossings are interpolated, and the collapse grid step is 0.005. The temperature grid has a node at $T = 2.269$, which essentially coincides with the exact T_c ; this can make argmax-plus-parabola and interpolated read-outs look better than they would on a grid offset from T_c , and we did not repeat the study on an offset grid. The sampler was checked against Onsager’s energy at four temperatures, not against critical observables, and decorrelation was checked through the autocorrelation of $|m|$ at one size only.
- Mechanisms. Only the symmetry explanation for the raw logistic regression is isolated by a control. Explanations for the size-independent bias of the CNN, for the failure of confusion collapses and for the biased CNN exponent are conjectures, not tested. The sweeps change one factor at a time and include only the MLP and CNN; the scaled-epoch training-size sweep matches the number of updates but not the learning-rate schedule or the total number of distinct configurations, and the remaining MLP sensitivity is not explained by it.
- Same-chain structure. The held-out chain is independent of the training chains, but the four chains per temperature share the sampler and, per seed, the random stream; independent MC runs with different algorithms would be a stronger test.

VIII. Conclusion

On identical Swendsen–Wang data for the 2D Ising model, a small CNN and an MLP trained only far from the transition give single-size 0.5-crossing estimates of T_c within a fraction of the temperature grid step at $L = 32$ (CNN 0.023, MLP 0.020), better than a PCA susceptibility read-out (0.056) and not distinguishable (Wilcoxon $p = 0.11$ for the CNN) from the central peak of learning by confusion, whose mean error is nominally the lowest at $L = 32$ (0.015). Logistic regression on raw spins is blind to the order because of the spin-flip symmetry, and gauge fixing, the control that isolates this, recovers only part of the performance. An FSS collapse of the classifier output did not improve the estimate here, the CNN collapse exponent was biased, and the confusion collapse failed in all variants. The MLP estimate depends more on the training-set size than on the excluded window, and this is not only a matter of the number of gradient updates; the CNN estimate does not. All of this holds for a small study (three sizes, ten seeds, untuned reimplemented baselines) and should be read as a measurement protocol and a set of negative results rather than a ranking of methods.

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